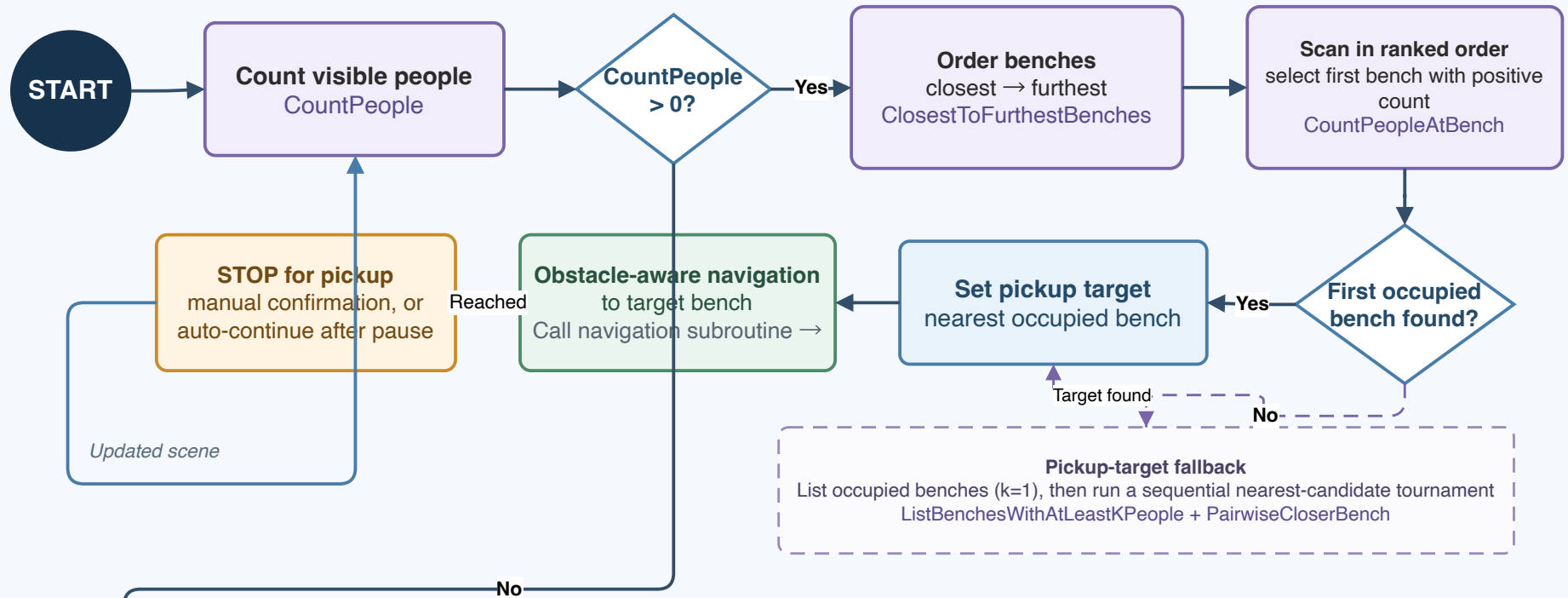


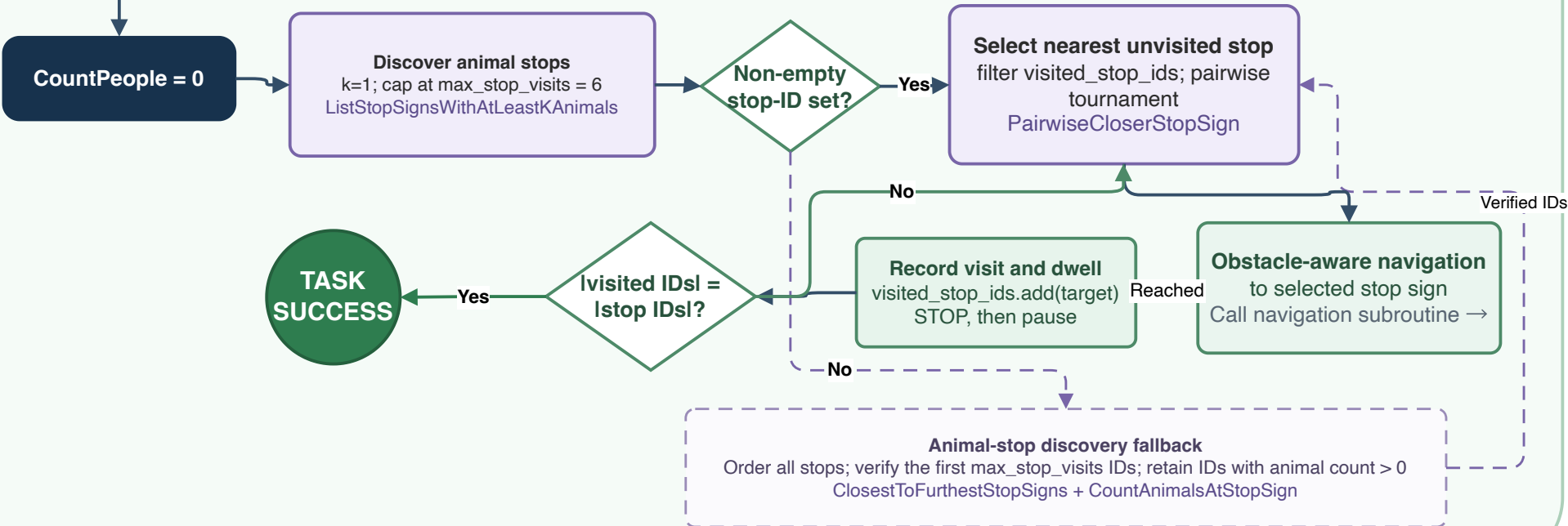
Zoo-Bus Deployment State Machine

Closed-loop VLM policy implemented in controlStack/zooBus.py

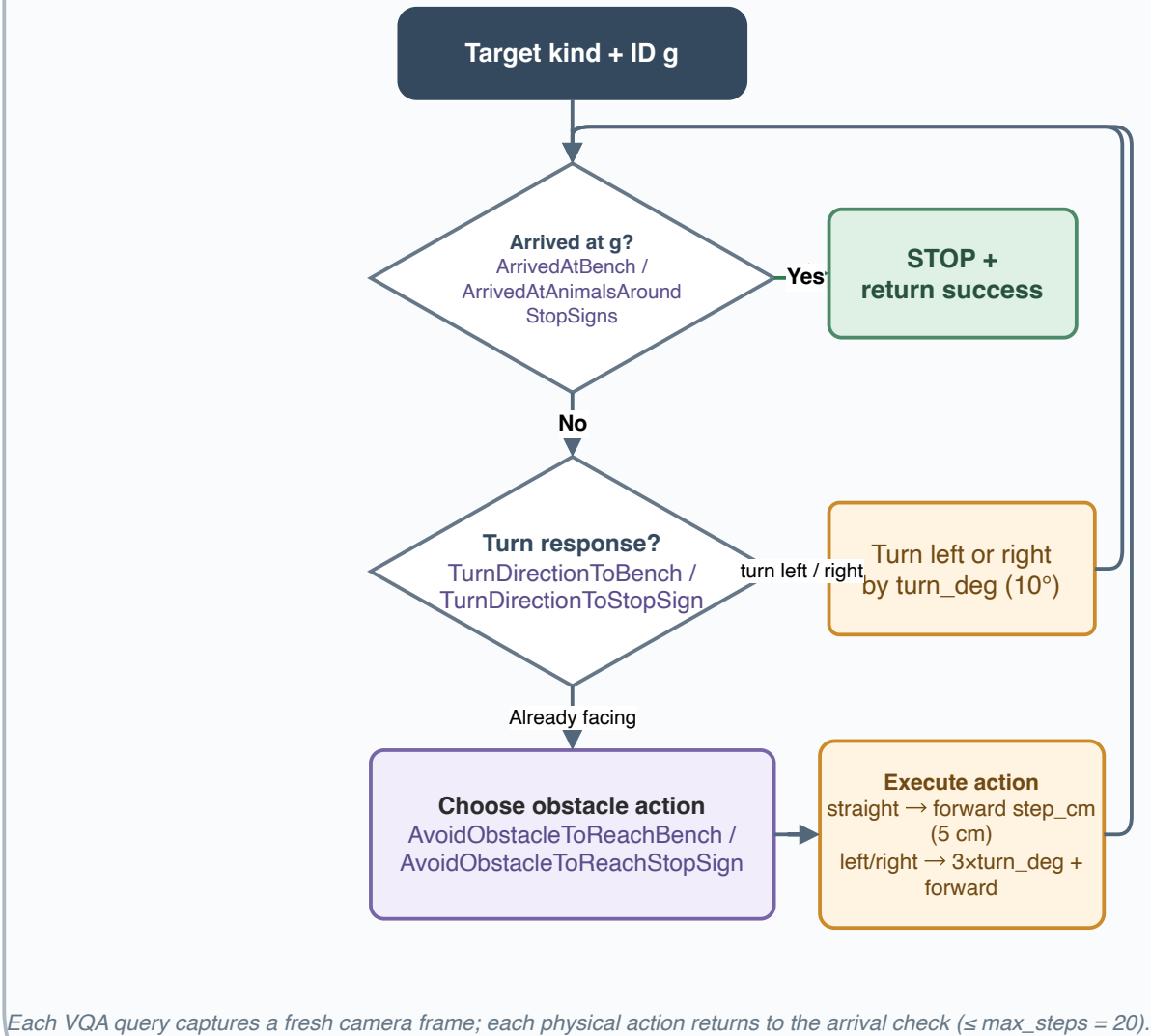
PHASE 1 · PICK UP ALL VISIBLE PASSENGERS



PHASE 2 · VISIT DISCOVERED ANIMAL STOP SIGNS WITHOUT REPETITION



SHARED CLOSED-LOOP OBSTACLE-AWARE NAVIGATION



Each VQA query captures a fresh camera frame; each physical action returns to the arrival check ($\leq \text{max_steps} = 20$).

Retries and termination guards
Validated list/ranking/count/selection/obstacle queries retry up to parse_retries + 1 attempts (default: 2); an invalid pairwise result leaves the incumbent winner unchanged. Arrival or turn parse failure aborts navigation.
Also fail on empty target sets, max_pickups = 8, navigation timeout, missing unvisited target, or manual interrupt. Camera and Bluetooth are closed in finally.

TASK FAILURE

Legend VQA query Decision Physical action Fallback / guard